

Networks

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Why Networking Matters

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Networking is the invisible fabric of the modern world. It's no longer just about computers; it's about *everything*.

- **Communication:** From email and social media to video calls.
- **Services:** Cloud computing, streaming (Netflix, Spotify), and online gaming.
- **Economy:** E-commerce, banking, and global financial systems.
- **IoT (Internet of Things):** Smart homes, wearable tech, and connected cars.

Understanding networking is no longer optional; it's a fundamental skill for any technologist.

Part 1: The Building Blocks

Key Network Equipment i

First, let's meet the hardware that builds a network.

- **Hub:** A “dumb” repeater. Any packet it receives is broadcast to *every* other port. It's slow, inefficient, and creates “collisions.” (Rarely used today).
- **Switch:** A “smart” device for a LAN. It learns which device is on which port (using MAC addresses) and sends packets *only* to the intended recipient.
- **Router:** A “gateway” that connects *different* networks. Your home router connects your private LAN to your provider's WAN (the Internet).
- **Access Point (AP):** A “translator” that connects wireless devices (using Wi-Fi) to the wired network (the switch).

- **ONT (Optical Network Terminal):** Your “modem” for a fiber-optic connection. It translates light signals from the fiber cable into electrical signals for your router (Ethernet).

Key Network Equipment iii

Common Types of Network Devices



Hub



Router



Gateway



NIC



Modem



Repeater



WAP



Firewall



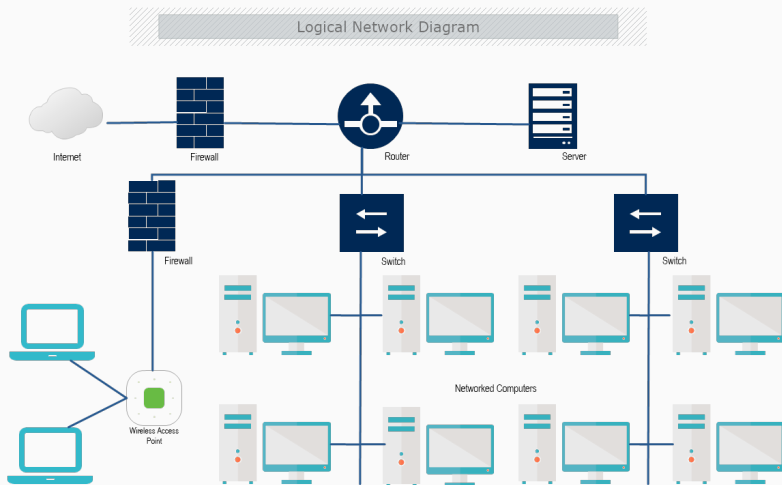
IDPS



VPN

Common Network Devices

Key Network Equipment Connected



Network Devices Connected

The Two-Address System: MAC & IP

Every device on a network has **two** addresses. Both are crucial.

- **MAC Address (Physical Address):**

- Example: 00:1A:2B:3C:4D:5E
- A unique, 48-bit serial number burned into the network card by the manufacturer. While intended to be permanent, it can be **spoofed** for privacy or security.
- **Used for:** Communication *within* the same Local Area Network (LAN).

- **IP Address (Logical Address):**

- Example: 192.168.1.10
- A logical, 32-bit (or 128-bit for IPv6) address assigned to the device by the network (e.g., by your router). It is temporary.
- **Used for:** Communication *between* different networks (on the WAN/Internet).

Analogy: A MAC address is like your **passport number** (permanent, identifies you). An IP address is like your **home**

How LANs *Really* Work: ARP

Your computer (192.168.1.10) wants to send a packet to your printer (192.168.1.15) on the same LAN.

- The **Router** only understands IP addresses.
- The **Switch** (which connects them) only understands MAC addresses.

How does the computer find the printer's MAC address?

1. It shouts to the whole LAN: "WHO HAS 192.168.1.15?"
This broadcast is the **Address Resolution Protocol (ARP)**.
2. The printer (192.168.1.15) replies: "I DO! My MAC is 00:AB:CD:EF:12:34."
3. Your computer stores this IP -> MAC pair in its **ARP table** and sends the packet.

The Language: IPv4 & The Subnet Mask

An IPv4 address alone isn't enough. It's always paired with a **Subnet Mask**.

- **IP Address:** 192.168.1.10
- **Subnet Mask:** 255.255.255.0

The subnet mask's job is to split the IP into two parts:

1. **Network Part:** 192.168.1.x ("What street am I on?")
2. **Host Part:** x.x.x.10 ("What is my house number?")

This is how your computer knows if another IP is **local** (on the same network) or **remote** (on a different network).

Special IPv4 Address Ranges

Not all IPs are equal. They are reserved for specific uses.

- **Loopback Address (localhost):**

- 127.0.0.1
- This address always means “**this computer.**” It’s a virtual interface used for testing applications on your own machine.

- **Private / LAN Addresses:**

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255
- These are for use *inside* a private network (LAN). They are not routable on the public internet.

- **Public / WAN Addresses:**

- Any other address (e.g., 8.8.8.8 or 142.250.184.142).
- These are globally unique and routable on the Internet.

Understanding IPv6 Addresses

IPv4 ran out of addresses. IPv6 is the successor and has its own special types.

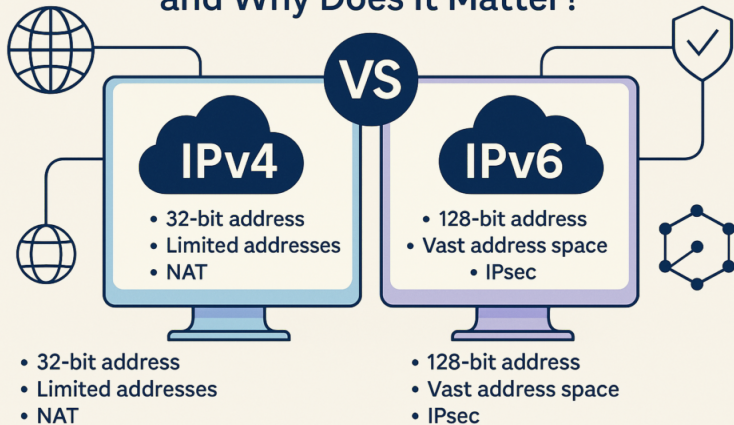
- **IPv6 (Internet Protocol v6):**

- A **128-bit** address (e.g.,
2001:0db8:85a3::8a2e:0370:7334).
- Provides a virtually limitless supply of addresses.

- **Special IPv6 Addresses:**

- **Loopback:** ::1 (The equivalent of 127.0.0.1).
- **Link-Local:** fe80::... (Automatically assigned for *local* LAN communication. Like ARP for IPv6).
- **Unique Local:** fd00::... (The equivalent of private IPv4 ranges).

IPv4 vs IPv6: What's the Difference and Why Does It Matter?



Addresses Aren't Enough: Ports

An IP address gets a data packet to the right *computer*. A **Port** gets it to the right *application* on that computer.

- **Analogy:** If an IP is the building's street address, the port is the apartment or office number.
- **Common Ports:**
 - 80: **HTTP** (Web)
 - 443: **HTTPS** (Secure Web)
 - 22: **SSH** (Secure Shell)

A connection is made to an **IP + Port** (e.g., 172.217.14.228:443).

Part 2: The Wide World (WAN)

Leaving the LAN: The Default Gateway

1. Your computer (192.168.1.10) wants to send a packet to Google (8.8.8.8).
2. It looks at its subnet mask (255 . 255 . 255 . 0).
3. It realizes that 8.8.8.8 is **not** on its local network.
4. It can't send the packet directly. So, it sends it to the **Default Gateway**.

The **Default Gateway** is the IP address of the **Router** on the LAN (e.g., 192 . 168 . 1 . 1). It's the "door" out of your local network, responsible for forwarding all non-local traffic.

Once the packet reaches your router, what's next?

- The **WAN (Wide Area Network)** is a network of networks (the Internet!).
- **Routing** is the process of finding the best path for data packets to travel from their source to their destination, hopping between thousands of different routers across the globe.

The “Receptionist”: NAT

NAT (Network Address Translation) is the clever workaround for the IPv4 address shortage.

- It allows an entire private network (e.g., all 50 devices in your home with 192.168.1.x addresses) to “hide” behind **one single public IP address**.
- Your router acts as a “receptionist,” keeping track of all outgoing requests and ensuring the responses get back to the correct private device.

Try using the [IPinfo](#) to identify your public IP.

The “Phonebook”: DNS

We remember names (google.com), but computers only understand numbers (142.250.184.142).

DNS (Domain Name System) is the “phonebook of the Internet.” It is a global, distributed system that translates human-readable domain names into machine-readable IP addresses.

More DNS Magic: mDNS & DDNS

- **mDNS (Multicast DNS):**

- This is “local” DNS. It lets devices on your LAN find each other by name *without* a central DNS server.
- This is how `my-laptop.local` or your printer automatically “appears.”
- Implemented by services like **Avahi** (Linux) or **Bonjour** (macOS/Windows).

- **DDNS (Dynamic DNS):**

- Your home’s public IP address can change (it’s “dynamic”).
- DDNS is a service that automatically updates a domain name to point to your new IP address whenever it changes.
- This is useful for hosting a server (e.g., Nextcloud) at home.

Keeping the Network Running: NTP & SNMP

- **NTP (Network Time Protocol):**
 - Keeps the clocks on all computers and network devices in sync.
 - This is **critical** for security (encryption certificates), financial transactions, and accurate log files.
- **SNMP (Simple Network Management Protocol):**
 - Used by network administrators to monitor the health, performance, and configuration of routers, switches, and servers.

Part 3: Application Protocols

Protocols are the “rules of conversation” for specific tasks.

- **HTTP (HyperText Transfer Protocol):** The fundamental protocol for the World Wide Web. It’s how your browser *requests* web pages.
- **HTTPS (HTTP Secure):** This is just HTTP layered on top of **SSL/TLS** encryption. It ensures your communication is private and secure. **Always look for the !**

- **SMTP (Simple Mail Transfer Protocol):**
 - Used for *sending* email.
- **POP3 (Post Office Protocol):**
 - Used for *retrieving* email.
 - It *downloads* mail to your device and (usually) deletes it from the server. This is an older model.
- **IMAP (Internet Message Access Protocol):**
 - Used for *retrieving* email.
 - It *syncs* your mail with the server. This is the modern model. What you do on your phone appears on your laptop.

- **SSH (Secure Shell):**
 - The single most important tool for system administrators.
 - Provides an encrypted command-line connection to a remote server.
- **FTP (File Transfer Protocol):**
 - An old, *insecure* (plain text) protocol for transferring files. **Avoid it.**
 - Use **SFTP** (which runs over SSH) instead.
- **WebDAV / CalDAV / CardDAV:**
 - Extensions of HTTP that let you manage files (WebDAV), calendars (CalDAV), and contacts (CardDAV) on a web server. Used by services like Nextcloud.

- **MQTT (Message Queuing Telemetry Transport):**
 - A very lightweight and efficient protocol designed for “publish” and “subscribe” messages (a pub/sub model).
 - Perfect for **IoT (Internet of Things)**: sensors, smart bulbs, and small devices that need to send tiny messages reliably with low power.

Part 4: Managing & Diagnosing

How does your device get an IP address?

- **Windows:**
 - Managed via the **Control Panel** or the Settings app.
- **Linux (Desktop):**
 - Almost always managed by **NetworkManager**, a user-friendly service with a graphical front-end (your network icon).
- **Linux (Server):**
 - Often managed by **systemd-networkd**.
 - Configuration is done via simple text files in `/etc/systemd/network/`.

Automatic Configuration: DHCP

In the last slide, we asked: “How does your device get an IP address?” For 99% of devices, the answer is **DHCP (Dynamic Host Configuration Protocol)**.

Manually setting an IP on every phone, laptop, and smart TV (a **static IP**) would be a nightmare. DHCP automates this.

1. Your device joins a network and shouts a **DHCP Discover** message: “Is there a DHCP server out there? I need an IP!”
2. A **DHCP Server** (usually your router) replies with a **DHCP Offer**: “Here, you can *use* 192.168.1.50.”
3. Your device accepts with a **DHCP Request**, and the server confirms with a **DHCP ACK** (Acknowledgement).

DHCP Leases & IP Ranges

DHCP doesn't give you an IP forever. It gives you a **lease**.

- **Lease Time:** The IP is “rented” to your device for a specific time (e.g., 24 hours). Before it expires, your device must renew the lease. This ensures that IPs from devices that leave the network are eventually returned to the pool.
- **IP Range (Pool):** The DHCP server is configured to manage a *range* of addresses (e.g., 192.168.1.100 to 192.168.1.200).
- **Static Attribution:** By only using a range, the server leaves other IPs free (e.g., 192.168.1.1 to 192.168.1.99) for **static assignment**. These are manually configured on important devices like servers, printers, and the router itself, so their addresses never change.

Diagnostic Tool 1: ping

- **The Question:** "Are you there?"
- **The Action:** Sends a small packet (ICMP Echo Request) and waits for a reply.
- **The Answer:** Tells you if a host is reachable and how long the round-trip took (this is **latency**).
- **Example:** ping google.com

Diagnostic Tool 2: traceroute

- **The Question:** “What path do my packets take to get to you?”
- **The Action:** Sends packets with increasing “Time-To-Live” (TTL) values.
- **The Answer:** Shows you every single router (or “hop”) your packet passes through on its way to the destination. Great for finding *where* a connection is failing.
- **Example:** `traceroute google.com`

Diagnostic Tool 3: ip, dig, nmap

- **ip addr show**
 - The modern Linux tool to view your *own* IP configuration and network interfaces. (Replaces the old `ifconfig`).
- **dig google.com**
 - **The Question:** “What is the IP address for this name?”
 - **The Action:** Performs a DNS lookup.
- **nmap localhost**
 - **The Question:** “What doors are open on this machine?”
 - **The Action:** A powerful port scanner.
 - **The Answer:** Reports which ports are open and what services are (likely) running on them.

Part 5: Security & Advanced Topics

Network Monitoring: Wireshark

- **The Tool:** Wireshark is a “network sniffer” or protocol analyzer.
- **The Action:** It captures *every single packet* traveling on your network interface and lets you inspect its contents.
- **Analogy:** It’s like a video camera for your network traffic.
- **Use:** The single most powerful tool for debugging complex network problems.

A **firewall** is the “security guard” for your network or computer.

- It inspects all incoming and outgoing network traffic.
- It decides whether to **allow** or **block** each packet based on a set of rules (e.g., “Allow traffic on port 443, block everything else”).
- This is your first line of defense.

Firewall Examples

- **iptables:** The classic, powerful, command-line firewall built into the Linux kernel for decades.
- **nftables:** The modern successor to iptables in Linux. It has a simpler syntax and better performance.
- **pfSense:** A free, open-source **firewall operating system**. You install it on a dedicated computer to turn it into an extremely powerful, enterprise-grade router and firewall for your entire network.

The Power of SSH: VS Code Remote

SSH is more than just a remote shell. The “Remote - SSH” extension in VS Code is a game-changer.

- **How it works:** Your editor UI runs locally, but all file editing, terminal commands, and language processing run on the remote server.
- You get the power of a server with the comfort of your local editor.

The Power of SSH: Tunnels & X11

- **SSH Tunnels (Port Forwarding):**
 - Lets you securely “wrap” network traffic inside an SSH connection.
 - **Example:** Access a database running on `localhost:5432` on a remote server as if it were running on *your own* machine's `localhost:5432`.
- **X11 Forwarding:**
 - Lets you run a *graphical* application (like `firefox` or a text editor) on a remote Linux server, but see and interact with the window on your local desktop.

Syncing Files: rsync over SSH

The best way to transfer and sync files. rsync is fast, efficient, and versatile.

- **Why it's fast:** It only copies the **differences** (deltas) between files, not the whole file.
- It works perfectly over an SSH connection.
- **Command:**

```
# Sync a local folder UP to a remote server  
$ rsync -avzP ./my-project/ user@host:~/projects/
```

The Modern Web: Reverse Proxy

A **Reverse Proxy** (like NGINX or Caddy) is a server that sits *in front* of your actual application servers.

- It receives all incoming traffic from the Internet.
- It then “proxies” (forwards) the request to the correct internal application (e.g., your Python app, your Node.js app).
- **Uses:**
 - **Load Balancing:** Distribute traffic across multiple app servers.
 - **Security:** Hides your application servers from the Internet.
 - **Hosting:** Host multiple websites on a single IP address.

The Modern Web: HTTPS & Let's Encrypt

- **HTTPS** is essential. It provides the encryption (SSL/TLS) that keeps user data private and proves your site's identity.
- **The Problem:** Certificates used to be expensive and difficult to install.
- **The Solution: Let's Encrypt**
 - A free, automated, and open **Certificate Authority (CA)**.
 - It provides free SSL/TLS certificates and tools (like certbot) to automatically install and renew them.
 - It has made the entire web more secure.

Further Reading & Resources

Further Reading & Resources

- **Wireshark:** <https://www.wireshark.org/>
- **Nmap:** <https://nmap.org/>
- **Let's Encrypt:** <https://letsencrypt.org/>
- **ip command guide:** <https://www.geeksforgeeks.org/ip-command-in-linux-with-examples/>
- **rsync guide:**
<https://www.digitalocean.com/community/tutorials/how-to-use-rsync-to-sync-local-and-remote-directories>
- **Mozilla's Guide to HTTP:**
<https://developer.mozilla.org/en-US/docs/Web/HTTP/Overview>